CITS4403 Project: Modelling the Perth Housing Crisis

Lauren Pudney

October 4, 2026

1 Background

Housing instablity exists as a critical socioeconomic issue within Australian society. Despite high levels of economic prosperity, escalating housing pressures and persistent homelessness continue to affect Australians [1, 2], placing disproportionate burden on marginalised and lower-income groups, such as young people, migrants and renters [4].

Over the past two decades, housing costs in Australia have increased significantly relative to incomes, with home-ownership for all age groups under 65 decreasing [4].

2 Reaearch aims

2.1 Research hypothesis

2.2 Contributions

3 Prior work

Data-Driven Modelling of Migration, Income Trends, and Rental Pressures to Design Targeted Interventions Reducing Housing Instability [6]
Modelling housing need in Australia to 2025 [7]
London's Housing Crisis; A System Dynamics Analysis of Long-term Developments: 40 Years into the Past and 40 Years into the Future [5]
HousingABM Japan: An Agent-Based Model of Price and Rent Amplification in the Tokyo Housing Market
https://www.comses.net/codebases/0fbece58-7669-4204-870f-1643e7821fb5/releases/1.0.0/ (not peer reviewed)

Model that our base model was inspired by:
General Housing Model [3]

4 Model specification

4.1 Assumptions

5 Methodology

6 Results

7 Discussion

8 Conclusion

9 Future work

9.0.1 Credit statement

Pudney: Core + solver backend implementation, benchmarking and survey design and implementation, document preparation. *Grinbergs*: loss backend implementation, documentation preparation. *Daggitt*: Conceptualisation, ITP

backend and documentation preparation.

References

[1] Estimating Homelessness: Census, 2021 — abs.gov.au. https://www.abs.gov.au/statistics/people/housing/estimating-homelessness-census/latest-release, [Accessed 04-10-2026]

[2] Aihw: Australia's welfare 2023: data insights. Tech. rep., AIHW, Australian Government (2023)

[3] Applegate, J.M.: General housing model (2020). https://doi.org/10.25937/v8yx-4g98

[4] Daley, J., Coates, B., Wiltshire, T.: Housing affordability: re-imagining the Australian dream. Grattan Institute (2018)

[5] Dianati, K.: London's Housing Crisis; A System Dynamics Analysis of Long-term Developments: 40 Years into the Past and 40 Years into the Future. Ph.D. thesis (04 2022)

[6] Matthew, D.: Data-driven modelling of migration, income trends, and rental pressures to design targeted interventions reducing housing instability (12 2025). https://doi.org/10.13140/RG.2.2.12720.83201

[7] Rowley, S., Leishman, C., Baker, E., Bentley, R., Lester, L.: Modelling housing need in australia to 2025. Tech. rep. (Jul 2017)